AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 1 · STUDENT LESSON

Representing inputs as vectors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Name the fields. Keep their order. Preserve their meaning.

Encode clear numerical inputs while distinguishing examples,features and unavailable targets.

Keep one habit: Parcel schema → unit conversion → brightness position picture → complete encoder probes → independent battery schema.

MAIN LESSON · SECTION 1

What does each number mean,and where does it belong?

> Keep names,units and positions together before doing network arithmetic.

Learning goals

I can encode a record using a fixed feature order and units.

I can distinguish a **vector dimension** from the number of examples.

I can detect missing values and target leakage before computation.

Retrieve the evidence habit

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Check the prediction-time target

- A parcel classifier uses width and height to predict a class. Why must its final class label stay outside the **input** features?

> **My starting explanation**

End goal

By the end, I can explain representing **input**s as **vector**s using a traced example and a stated limitation.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Encode a record using a fixed feature order and units

Distinguish a **vector dimension** from the number of examples

Detect missing values and target leakage before computation

Retrieve what you already know

This week’s end goal

MAIN LESSON · SECTION 2

One example is one ordered vector

> Changing a coordinate changes the meaning of the input.

Declare the order,units and valid measurements

A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields.

A **vector** is an ordered list: [4, 2] can mean width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same **schema**.

For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The **vector** has **dimension** 2 because it contains two **component**s.

Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

Three complete parcel records

Record	Width and height in cm	Input vector
A	4 and 2	[4, 2]
B	2 and 4	[2, 4]

Record	Width and height in cm	Input vector
C	3 and 3	[3, 3]

Separate feature dimension from example count

There are three examples, each with two features. Stacking their **vector**s makes a table with three rows and two columns. Do not confuse the number of rows with the **dimension** of one **input**.

Name supplied and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Read names,convert units,validate,then encode

> The encoder prepares an input;it does not train a model or predict a class.

Complete step-by-step conversion

Encode parcel D	Value / check
Read named source fields	width 50 mm, height 30 mm
Convert each measurement to cm	width 50÷10=5, height 30÷10=3

Encode parcel D	Value / check
Check the declared input contract	Both measurements present and positive
Write the fixed coordinate order	[width_cm, height_cm]=[5, 3]
Exclude audit metadata and target	Record ID and future class are not coordinates

Read fields by name, convert to the declared units, validate, and only then write the **vector**. Parcel D has width 50 mm and height 30 mm. Because 10 mm equals 1 cm, its encoded **vector** is [5, 3].

Writing [50, 30] would preserve the numbers while changing the physical meaning by a factor of ten.

Runnable encoder and its input assumptions

```
def encode(width_mm, height_mm):
    if width_mm is None or height_mm is None:
        raise ValueError("Missing measurement")
    if width_mm <= 0 or height_mm <= 0:
        raise ValueError("Measurements must be positive")
    return [width_mm / 10, height_mm / 10]

print(encode(50, 30))  # [5.0, 3.0]
```

This short function assumes each supplied value is a number in mm or None for missing. It is a classroom example, not a complete **input** validator. Trace the conversion before running it.

The function represents data; it does not fit a model or predict a class.

Connect each picture step to a named field

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare a unit change with an order change

> Explain which part of the input contract changed.

Try a new parcel measurement

- A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured coordinate swap

- Change: Swap only A's **vector** order before **prediction**.

▼ Check after predicting

Prediction: The encoded **input** becomes [7, 2], numerically identical to B's original **input**.

Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

▼ Earlier wording, alternatives and picture transcripts

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

See exactly what order preserves and a sum loses

> Use the same coordinate meaning for every example.

Complete brightness facts and schema boundary

Encode the same pixel measurements in a fixed **vector** order.

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Keep the two schemas separate:parcel coordinates mean width_cm/height_cm; picture coordinates mean left/right brightness on0–10. In the brightness schema, A=[2, 7] and B=[7, 2]. Both sum to9, but position 0 and position 1 represent different locations. Keeping the ordered pair preserves this distinction; keeping only the sum loses it. Swapping only A's order makes its supplied numbers match B's original vector.

That is an encoding change, not evidence that a network has learned vision.

Trace the ordered coordinates

AI ACADEMY / AI-5/1 · 02 · TRACE THE CASE

Follow the deciding steps

01 Declare coordinate order

Coordinate 0 is left brightness; coordinate 1 is right brightness.

02 Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.

03 Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coordinate order	Coordinate 0 is left brightness; coordinate 1 is right brightness.
2. Encode both images	A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.
3. Check input contract	Use the same order when training and predicting; record missing measurements rather than guessing.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Explain the information that remains missing

- These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

- Read the labelled picture: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Predict the swapped input

- Change one thing: Swap only A's **vector** order before **prediction**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-5/1 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Use the same order when training and predicting; record missing measurements rather than guessing.

Changed condition

Swap only A's vector order before prediction.

Worked changed-case result

The encoded input becomes [7, 2], numerically identical to B's original input.
Why: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Follow the labelled picture with the matching explanation. Open the vector diagram

Return to the parcel conversion and E attempt

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.

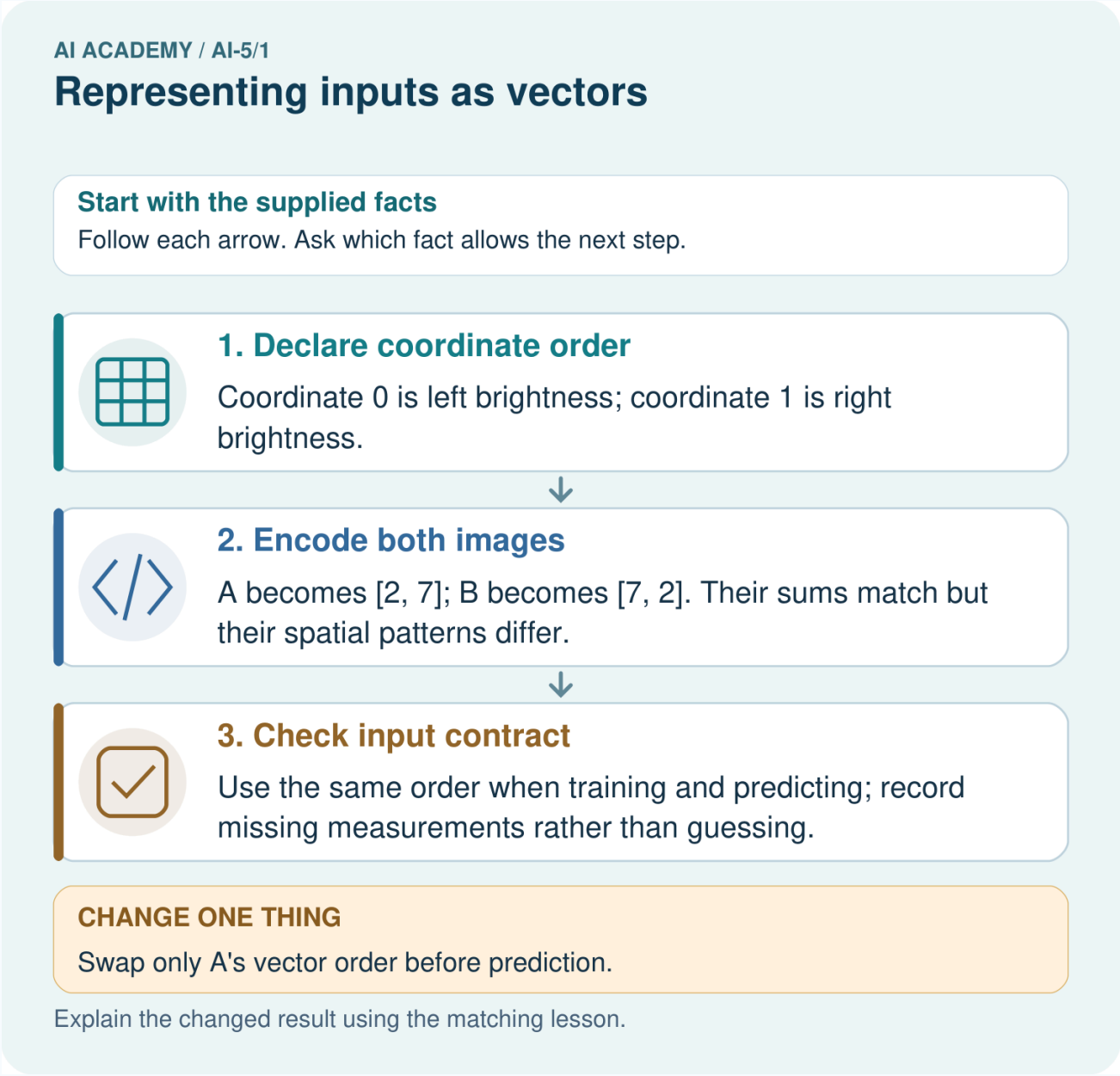
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Representing inputs as vectors

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Find the first invalid field or assumption

> A vector can have the right length and the wrong meaning.

Missing measurements,coordinate order and leakage

A missing height is unknown, not zero. Under this **schema** zero is also invalid, so reject the incomplete record and request the measurement.

A different project might use a documented **training**-fitted replacement plus a missingness flag; that would be a different **schema** that must be applied consistently.

A changed field order must also change every consumer that reads positions. A **vector** can have the expected length and still be wrong because its order, units or meaning changed.

A class label appended as a third **component** causes target leakage even if the software accepts three values.

Repair the incomplete parcel

- Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise the same schema with less help

> Each learner explains the named fields,not only the final numbers.

Encode A–C and try E

- Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

> **Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.**

> **Explain your prediction to a partner before comparing results.**

Use the supported W2record

Use workbook W2 for the connected example “Representing **input**s as **vector**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check whether order matters

- Use the worked case for Representing **input**s as **vector**s. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured distinction

- Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Convert before encoding

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate complete normal,boundary,missing and unit-error cases

> A numerical result still needs the correct field meanings.

Complete shared task and runnable encoder

A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

```
def encode(width_mm, height_mm):
    if width_mm is None or height_mm is None:
        raise ValueError("Missing measurement")
    if width_mm <= 0 or height_mm <= 0:
        raise ValueError("Measurements must be positive")
    return [width_mm / 10, height_mm / 10]

print(encode(50, 30))  # [5.0, 3.0]
```

All supplied probes and expected behavior

Complete encoder probe; inputs in mm	Expected output / action
Normal width 60, height 10	[6.0, 1.0] in cm

Complete encoder probe; inputs in mm	Expected output / action
Small positive boundary width 1, height 1	[0.1, 0.1]; positive small measurements are valid
Missing width 40, heightNone	ValueError:Missing measurement; request the unknown measurement
Zero width 40, height 0	ValueError:Measurements must be positive under this parcel schema
Changed unit:already-cm values 6, 1 passed as mm	Produces[0.6, 0.1], a tenfold under-encoding; fix the named source unit contract, not the output by guessing
Metadata or future class supplied	Keep outside these two coordinates; the encoder has no target argument

Predict,run or trace,and explain

1. Predict every probe before running or tracing it. Copy the supplied encode function and call it once per case; use separate runs for failing inputs so one error does not stop the rest. Save the actual output/error or mark paper trace. Then explain why the same zero can be invalid for parcel height but valid for a different declared field such as age.

   This function assumes numeric mm values or None; it is not a general parser for arbitrary strings or nonfinite numbers.

2. On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Keep the evidence ledger

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: encode a new battery schema

> Use the newly supplied field order and validity rules,not the parcel rules.

Complete new independent schema and records

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Use feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new schema independently

Attempt this without hints

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Start the project with a readable input contract

> Show the source field,unit,coordinate and missing-input rule.

Current project artifact

Add a labelled representing **input**s as **vector**s **evidence** sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Keep the actual evidence and scope

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish by naming each coordinate and one limit

> Retrieve before returning to the worked example.

Explain without notes

- Close your notes. Explain this goal in your own words: Encode a record using a fixed feature order and units

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Plan the changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Vector	An ordered list of numerical components
Component	One value at a specified position in a vector
Dimension	The number of components in a vector
Schema	The recorded feature order, units and validity rules

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.
- Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.
- Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new battery schema and K/Lrecords for the first individual attempt.

Current explanation and response

- Two parcels can share width and height yet differ in material or weight. Their **vector**s are identical in this representation, so a deterministic model receiving only these features cannot distinguish them.

  Adding a feature changes the **schema** and **dimension**; it does not automatically improve **prediction**s. Keep data splits and evaluation discipline from Level 4 (Grade 9).

- Next week, we will multiply each **component** by a matching weight and add the results. The computation is meaningful only if the feature order stays fixed.

- Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

> **Two records both encode as [4, 2] but have different target classes. What can this representation reveal, and what can it not reveal?**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ Reference · The earlier route checklist

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Encode the same pixel measurements in a fixed **vector** order.

By the end, I can	How I will show it
Encode a record using a fixed feature order and units	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a vector dimension from the number of examples	Explain the deciding step, show a trace or test, and state one limit.
Detect missing values and target leakage before computation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Encode the same pixel measurements in a fixed **vector** order.
- Which labelled step matches this detail? Use the same order when **training** and predicting; record missing measurements rather than guessing.
- 1. Declare coordinate order
- 2. Check **input** contract
- 3. Encode both images
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Check **input** contract Use the same order when **training** and predicting; record missing measurements rather than guessing.

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Changed-case question and optional hint

- Changed-case practice: Swap only A's **vector** order before **prediction**. Which conclusion is supported by the supplied facts?
- 1. The encoded **input** becomes [7, 2], numerically identical to B's original **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Repair and retain the record

Review ML Builder Expo (ai-4/36). Goal: Defend the project using traceable **evidence**. Short practice: Which three supporting items should be visible at the expo?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

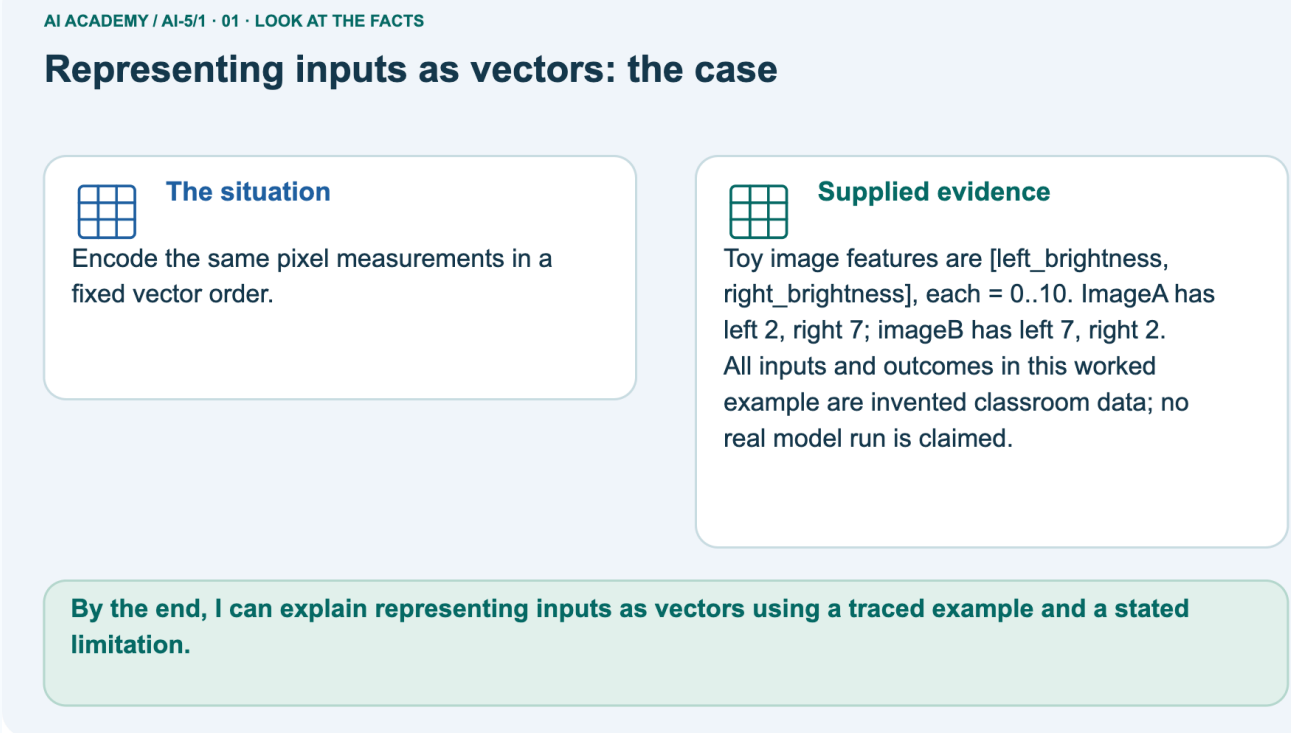
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Representing inputs as vectors: the case



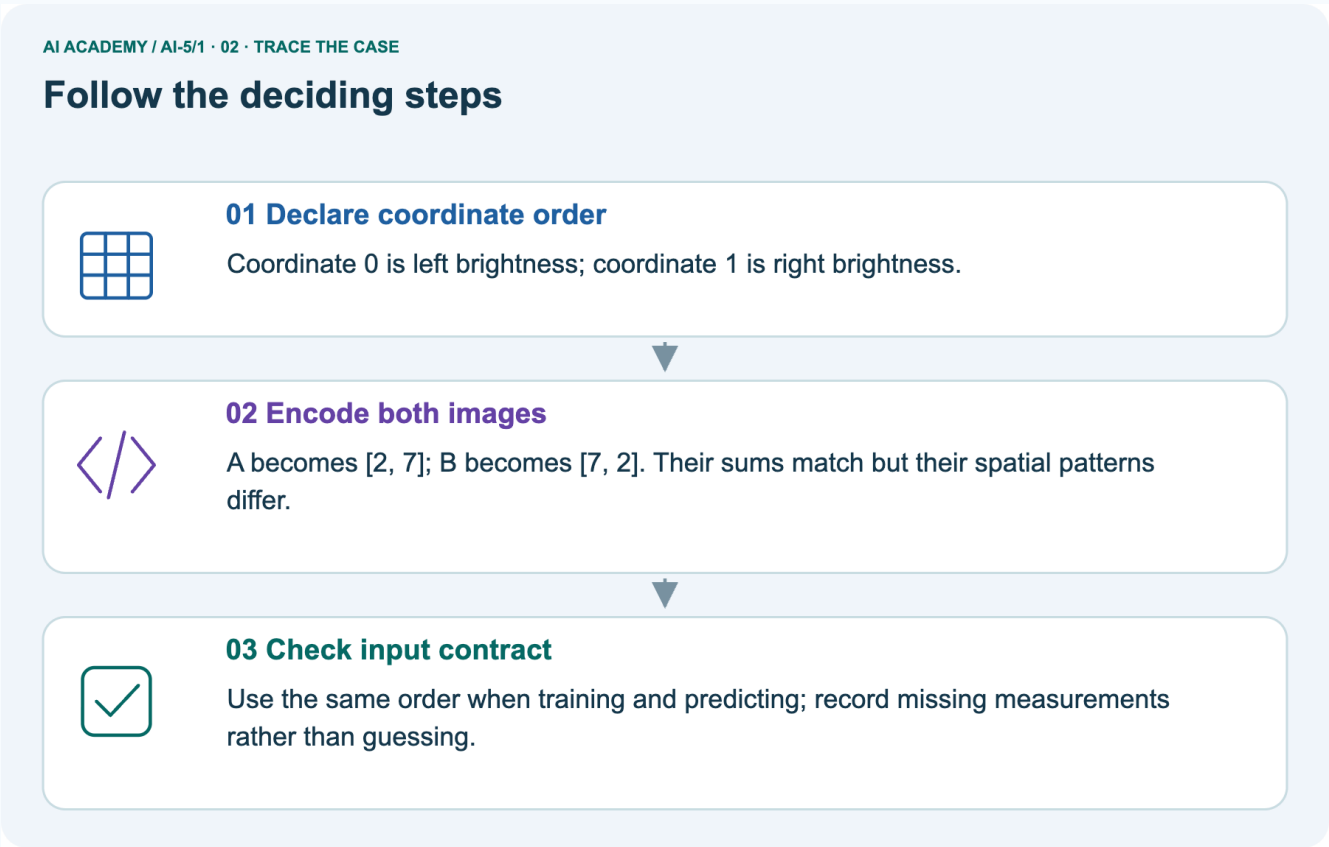
Follow the labelled picture with the matching explanation. Open the vector diagram

Encode the same pixel measurements in a fixed **vector** order. Supplied facts: Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



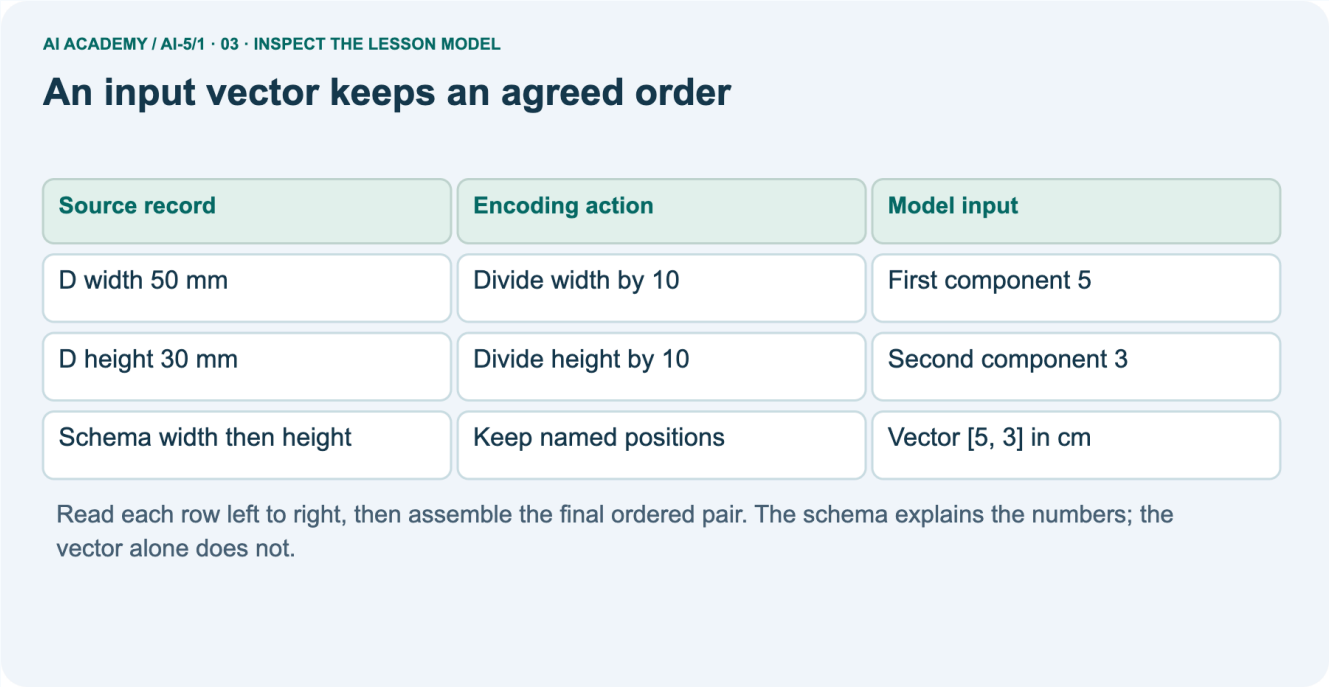
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coordinate order: Coordinate 0 is left brightness; coordinate 1 is right brightness. → Encode both images: A becomes [2, 7]; B becomes [7, 2].

Their sums match but their spatial patterns differ. → Check **input** contract: Use the same order when **training** and predicting; record missing measurements rather than guessing.

Point to each numbered stage and explain which recorded **input** it uses.

03. An input vector keeps an agreed order

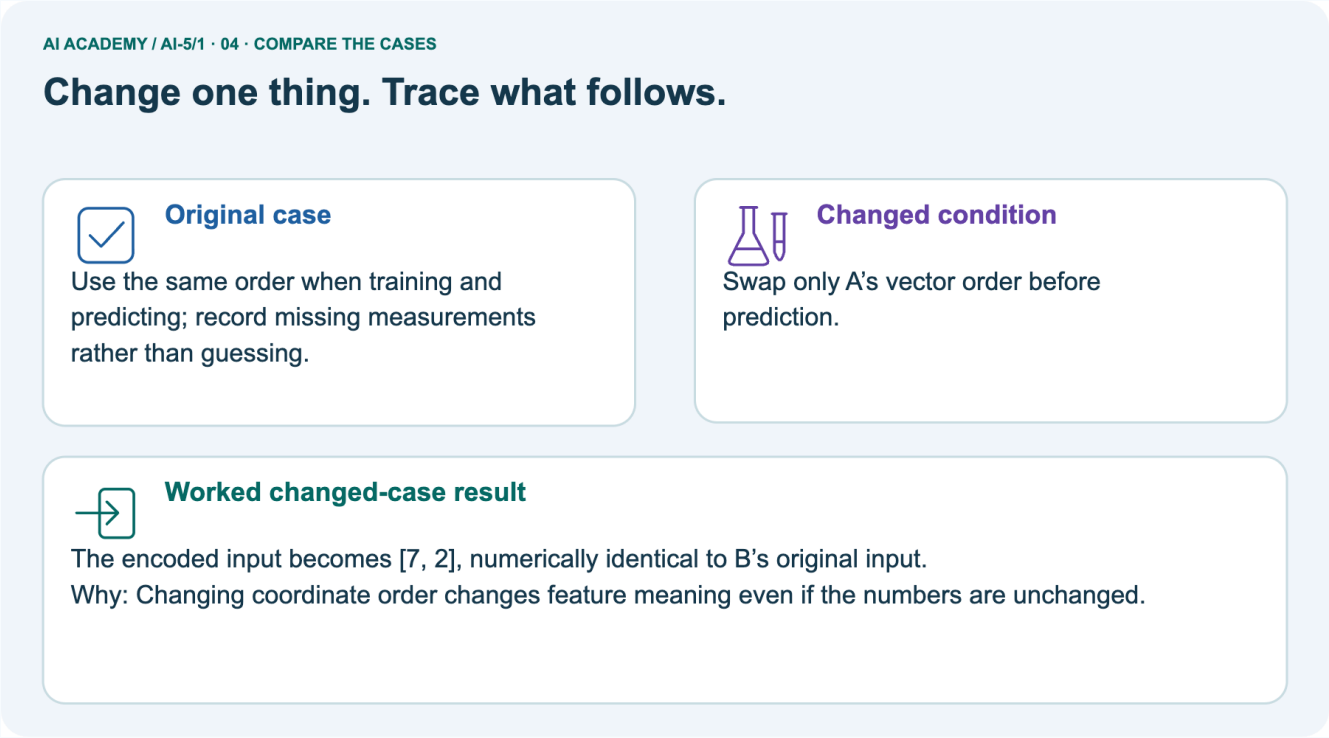


Follow the labelled picture with the matching explanation. Open the vector diagram

An **input vector** keeps an agreed order. D width 50 mm | Divide width by 10 | First **component** 5. D height 30 mm | Divide height by 10 | Second **component** 3. **Schema** width then height | Keep named positions | **Vector** [5, 3] in cm

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



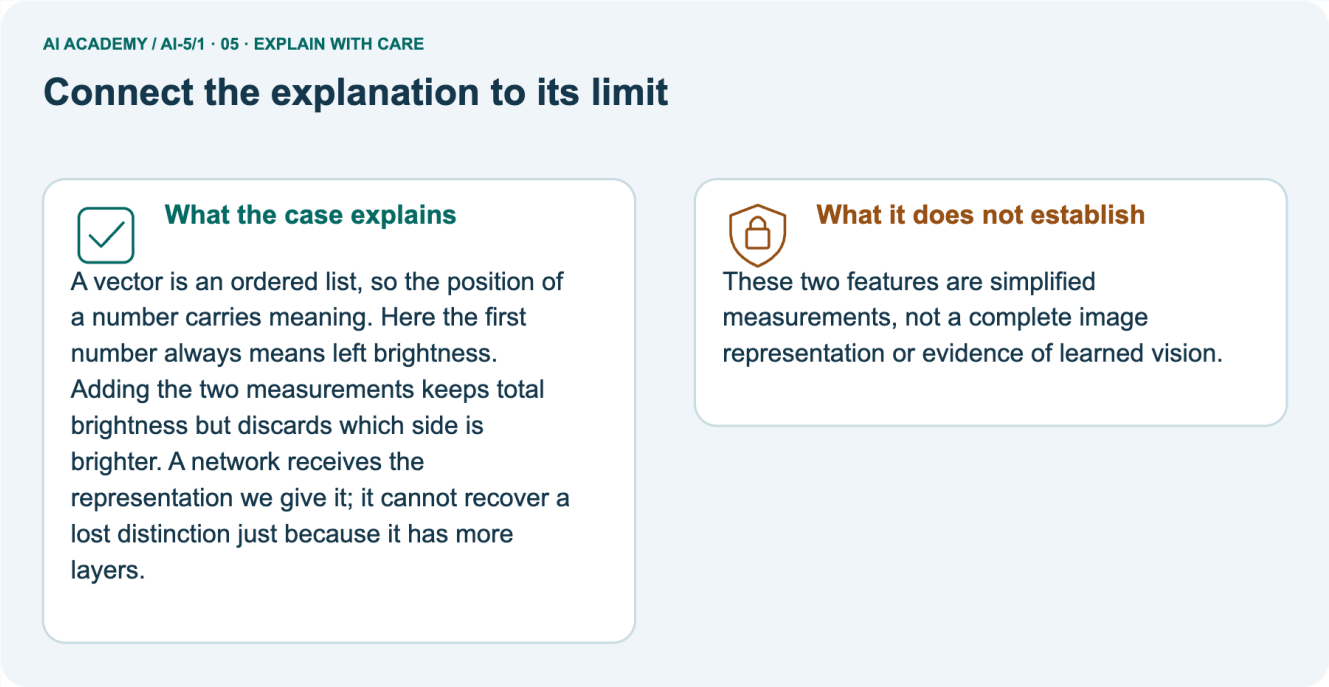
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use the same order when **training** and predicting; record missing measurements rather than guessing. Changed condition: Swap only A's **vector** order before **prediction**.

Expected result: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

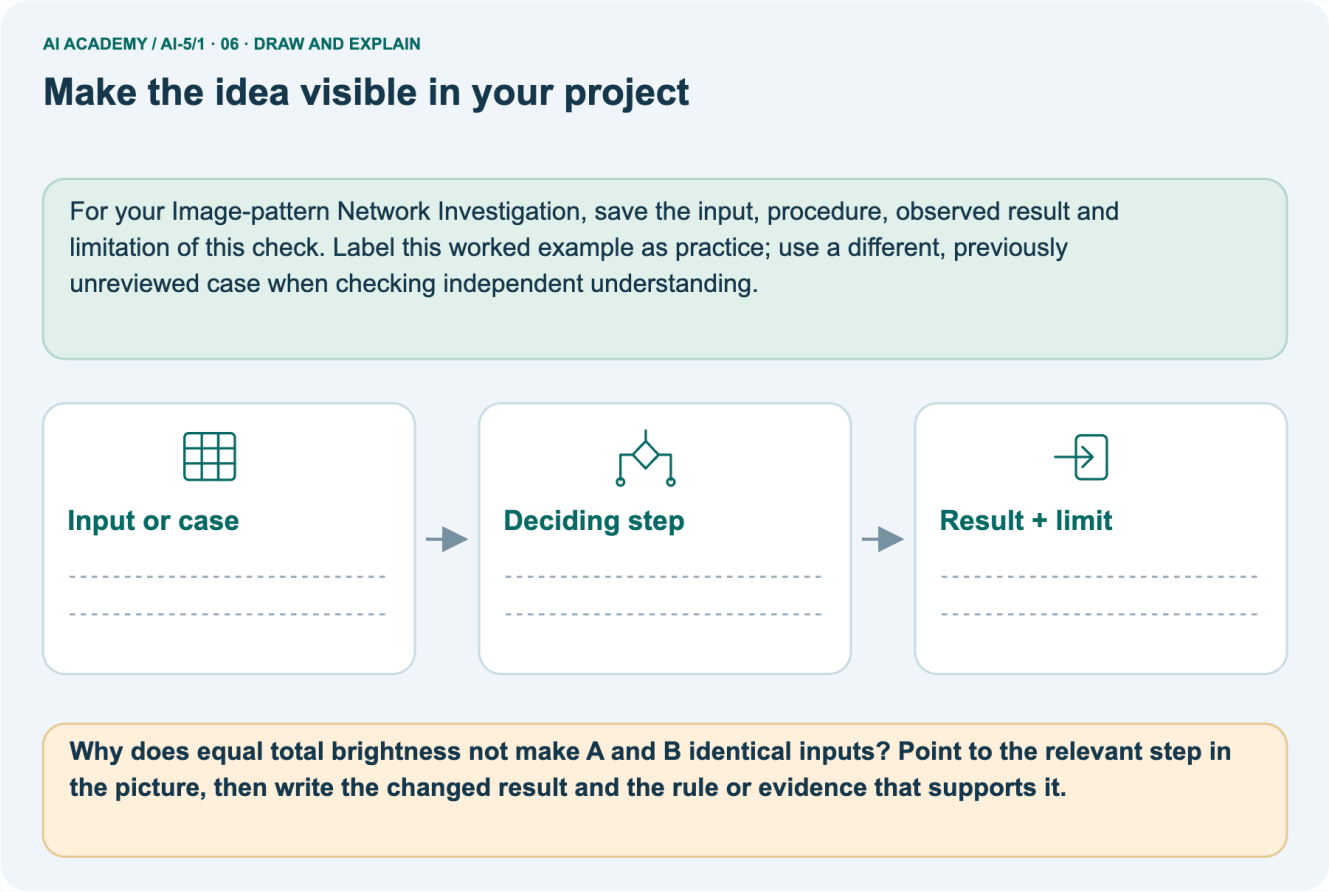
A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Limit: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Representing inputs as vectors: the case

02. Follow the deciding steps

03. An input vector keeps an agreed order

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.